

## CITIZEN RECORD INFORMATION SYSTEM

**Package:Com.Wipro.Citizen.Bean**

namespace Com.Wipro.Citizen.Bean

```
{  
    public class CitizenBean  
    {  
        public string citizenID;  
        public string citizenName;  
        public int age;  
        public string city;  
        public string status;  
  
        public string CitizenID  
        {  
            get { return citizenID; }  
            set { citizenID = value; }  
        }  
  
        public string CitizenName  
        {  
            get { return citizenName; }  
            set { citizenName = value; }  
        }  
  
        public int Age  
        {  
            get { return age; }  
            set { age = value; }  
        }  
    }  
}
```

```

    public string City
    {
        get { return city; }
        set { city = value; }
    }

    public string Status
    {
        get { return status; }
        set { status = value; }
    }
}

```

**Package: Com.Wipro.Citizen.Dao**

```

using Microsoft.Data.SqlClient;
using Com.Wipro.Citizen.Bean;
using Com.Wipro.Citizen.Util;

```

```

namespace Com.Wipro.Citizen.Dao
{
    public class CitizenDAO
    {
        public bool validateCitizen(string citizenID)
        {
            SqlConnection con = DBUtil.GetDBConnection();
            con.Open();

            SqlCommand cmd = new SqlCommand(
                "select count(*) from CITIZEN_TBL where Citizen_ID=@id", con);

```

```

cmd.Parameters.AddWithValue("@id", citizenID);

int count = (int)cmd.ExecuteScalar();
con.Close();

return count > 0;
}

public CitizenBean viewCitizen(string citizenID)
{
    CitizenBean cb = null;

    SqlConnection con = DBUtil.GetDBConnection();
    con.Open();

    SqlCommand cmd = new SqlCommand(
        "select * from CITIZEN_TBL where Citizen_ID=@id", con);

    cmd.Parameters.AddWithValue("@id", citizenID);

    SqlDataReader dr = cmd.ExecuteReader();

    if (dr.Read())
    {
        cb = new CitizenBean();
        cb.CitizenID = dr[0].ToString();
        cb.CitizenName = dr[1].ToString();
        cb.Age = int.Parse(dr[2].ToString());
        cb.City = dr[3].ToString();
        cb.Status = dr[4].ToString();
    }
}

```

```

    }

    con.Close();
    return cb;
}

public bool registerCitizen(CitizenBean citizenBean)
{
    SqlConnection con = DBUtil.GetDBConnection();
    con.Open();

    SqlCommand cmd = new SqlCommand(
        "insert into CITIZEN_TBL values(@id,@name,@age,@city,@status)", con);

    cmd.Parameters.AddWithValue("@id", citizenBean.CitizenID);
    cmd.Parameters.AddWithValue("@name", citizenBean.CitizenName);
    cmd.Parameters.AddWithValue("@age", citizenBean.Age);
    cmd.Parameters.AddWithValue("@city", citizenBean.City);
    cmd.Parameters.AddWithValue("@status", citizenBean.Status);

    int rows = cmd.ExecuteNonQuery();
    con.Close();

    return rows > 0;
}
}
}

```

**Package: Com.Wipro.Citizen.Util**

```
using Microsoft.Data.SqlClient;
```

```
namespace Com.Wipro.Citizen.Util
```

```
{
```

```
    public class DBUtil
```

```
    {
```

```
        public static SqlConnection GetDBConnection()
```

```
        {
```

```
            SqlConnection con = new SqlConnection(
```

```
                @"Data Source=localhost\SQLEXPRESS;Initial Catalog=CITIZENDB;Integrated  
Security=True;TrustServerCertificate=True");
```

```
            return con;
```

```
        }
```

```
    }
```

```
}
```

```
using System;
```

```
namespace Com.Wipro.Citizen.Util
```

```
{
```

```
    public class InvalidCitizenException : Exception
```

```
    {
```

```
        public override string ToString()
```

```
        {
```

```
            return "INVALID CITIZEN DETAILS";
```

```
        }
```

```
    }
```

```
}
```

**Package: Com.Wipro.Citizen.Service**

```
using System;
using Com.Wipro.Citizen.Bean;
using Com.Wipro.Citizen.Dao;
using Com.Wipro.Citizen.Util;

namespace Com.Wipro.Citizen.Service
{
    public class CitizenMain
    {
        CitizenDAO dao = new CitizenDAO();

        public string viewCitizen(string citizenID)
        {
            if (dao.validateCitizen(citizenID))
            {
                CitizenBean cb = dao.viewCitizen(citizenID);
                return "CITIZEN DETAILS FOUND";
            }
            return "CITIZEN NOT FOUND";
        }

        public string register(CitizenBean citizenBean)
        {
            if (citizenBean == null)
                return "INVALID";

            try
            {
                if (dao.validateCitizen(citizenBean.CitizenID) || citizenBean.Age < 18)
```

```

    {
        throw new InvalidCitizenException();
    }

    bool result = dao.registerCitizen(citizenBean);

    if (result)
        return "SUCCESS";
    }
    catch (InvalidCitizenException e)
    {
        return e.ToString();
    }

    return "INVALID";
}

public static void Main(string[] args)
{
    CitizenMain citizenMain = new CitizenMain();

    Console.WriteLine(citizenMain.viewCitizen("C10001"));

    CitizenBean c1 = new CitizenBean();
    c1.CitizenID = "C10008";
    c1.CitizenName = "rahul";
    c1.Age = 29;
    c1.City = "Madurai";
    c1.Status = "Active";
    Console.WriteLine(citizenMain.register(c1));
}

```

```

        CitizenBean c2 = new CitizenBean();

        c2.CitizenID = "C10004";

        c2.CitizenName = "Kavi";

        c2.Age = 16;

        c2.City = "Salem";

        c2.Status = "Active";

        Console.WriteLine(citizenMain.register(c2));


        Console.WriteLine(citizenMain.viewCitizen("C99999"));


        Console.WriteLine(citizenMain.register(null));
    }
}
}

```

Output:

```

Microsoft Visual Studio Debug Console
CITIZEN DETAILS FOUND
INVALID CITIZEN DETAILS
INVALID CITIZEN DETAILS
CITIZEN NOT FOUND
INVALID

C:\Users\praka\OneDrive\Desktop\dotnet tasks\CITIZENRECORD\CitizenRecord\bin\Debug\net10.0\CitizenRecord.exe (process 9604) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

## Database Design

### Table Name:CITIZENDB

```

CREATE TABLE CITIZEN_TBL (
    Citizen_ID VARCHAR(12) PRIMARY KEY,
    Citizen_Name VARCHAR(30),
    Age INT,
    City VARCHAR(30),

```



```
Status VARCHAR(10)

);

GO
```

```
INSERT INTO CITIZEN_TBL (Citizen_ID, Citizen_Name, Age, City, Status)

VALUES

('C10001', 'Ramesh', 35, 'Chennai', 'Active'),

('C10002', 'Sita', 28, 'Coimbatore', 'Active');

GO
```

```
INSERT INTO CITIZEN_TBL (Citizen_ID,Citizen_Name,Age,City,Status) VALUES

('C10003','sanjay',22,'Dindigul','Active');

GO
```

```
INSERT INTO CITIZEN_TBL (Citizen_ID,Citizen_Name,Age,City,Status) VALUES

('C10004','prakash',22,'Dindigul','Active');

GO
```

```
SELECT * FROM CITIZEN_TBL;

select count(*) from citizen_tbl where citizen_ID = 'C10002';
```

### Output:

| Results |            | Messages     |     |            |        |  |
|---------|------------|--------------|-----|------------|--------|--|
|         | Citizen_ID | Citizen_Name | Age | City       | Status |  |
| 1       | C10001     | Ramesh       | 35  | Chennai    | Active |  |
| 2       | C10002     | Sita         | 28  | Coimbatore | Active |  |
| 3       | C10003     | sanjay       | 22  | Dindigul   | active |  |
| 4       | C10004     | prakash      | 22  | Dindigul   | active |  |
| 5       | C10008     | rahul        | 29  | Madurai    | Active |  |
| 6       | C20001     | Arun         | 29  | Madurai    | Active |  |
| 7       | C30001     | dinakar      | 29  | sakthi     | Active |  |

